Biometric Authentication Mechanisms

Abstract

Biometric credentials are widely used in modern authentication: Sensors obtain a measurement and output *accept* if it is above a set match threshold or *reject* otherwise. A large body of work improves the accuracy of biometric sensors, but the state of the art uses a heuristic threshold. Other works study *authentication mechanisms*, but those treat biometric credentials as *atomic* (accept/reject, using the heuristic threshold).

We explore optimal authentication mechanisms with biometric credentials. Following prior work, we define a mechanism's success as the probability it accepts legitimate users and rejects attackers. We formally prove that in general the optimal threshold differs from the common heuristic. Moreover, a different optimum should be used according to the integration with other credentials. We also prove that integrating a biometric credential improves mechanism success even when an atomic credential has arbitrary low fault probabilities and a biometric credential provides only weak separation between users and attackers. We quantitatively demonstrate the improvement with experiments on real public biometric data. Our optimized threshold reduces mechanism failure probability by 30-70% compared to the heuristic threshold when a biometric credential is integrated with other credentials.

These results imply that biometric sensors should allow integrators to set thresholds according to the authentication mechanism.

CCS Concepts

• **Security and privacy → Formal methods and theory of security**; **Authentication**; **Biometrics**.

Keywords

Authentication Mechanisms, Biometrics

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1 Introduction

Biometric credentials are one of the most prevalent keys in modern authentication. Historically, they were used in signatures and handprints on royal documents and contracts, as they rely on unique physical characteristics of the user made them unforgeable. In the early days of the Internet, biometric credentials were not applicable and therefore *atomic credentials*, like physical devices or passwords, either a perfect match or wrong, were adopted. With the rapid growth of online services, more daily activities has transitioned to digital platforms. This expansion led companies to deploy advanced authentication mechanisms, such as multi-factor authentication (MFA) [3], increasing significantly the number of atomic credentials a user must have or remember to authenticate. However, password vulnerabilities persist: 59% of users in the U.S. reported leaks and 22% frequently forget their passwords, as of 2026 [11]. Consequently, biometric credentials are increasingly adopted, present in most modern user devices. The biometric sensors obtain a sample, calculate a similarity score between a sample and a reference stored in the database, and output accept if the score is above a set match threshold or reject otherwise.

Previous work (§2) has conducted extensive analysis of biometric credentials in isolation, primarily focusing on system accuracy and performance, aiming to reduce the false acceptance (FAR) and false rejection (FRR) rates [9, 18]. Sarkar et al. [21] introduce the Equal Error Rate (EER, where FAR = FRR) a heuristic operating point of biometric sensors. Hong et al. [8] propose a fused biometric system in which the final decision is based on the matching scores of both face and fingerprint. While these works widely cover biometric credentials, none of them analyze their integration into an authentication mechanism. Other works proposed authentication mechanisms defined as predicate of credential possession [3, 6, 13, 16], where each credential is treated as atomic. However, such settings are partially applicable to our framework. While we adopt their definition of *mechanism's success* as the probability it accepts users and rejects attackers, they treat biometric as atomic credntials. This leaves open the question of whether adapting biometric credentials in the presence of other credentials, can form the authentication mechanism to enhance its success.

In this work, we study how biometric credentials influence the authentication mechanism's success with and without atomic credentials. More precisely, our goal is to optimally use biometric credentials in authentication mechanisms and analyze how the presence of atomic credentials may adapt the biometric credential to enhace mehcanism's success. Fuse accepted models to form the authentication model (§3) built on a set of biometric and atomic credentials: A mechanism interacts with two players: a user and an attacker, trying to identify the user based on the provided credentials by the player. Each credential can be either [6], (1) *safe*: Only available to the user, (2) *lost*: Not available to either player, (3) *leaked*: Available to both players, or (4) *stolen*: Only available to the attacker. Each credential has a probability of being in each of the four states. Biometric credentials are evaluated by a sensor that calculates a confidence score for either player. If the score is above a *match threshold*, the mechanism approves an acess to the player, otherwise it denies. This is used to establish two fault probabilites: *false acceptance rate (FAR)*: The probability that the mechanism incorrectly accepts an attacker as a legitimate user, and *false rejection rate (FRR)*: The probability that the mechanism incorrectly rejects a legitimate user.

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We start by optimizing a single biometirc credential mechnaism (§4). We examine distribution functions that approximate a biometric credential score distribution 1. We find closed-form expressions of the mechanism's success for different score distribution functions of a biometric credential and we symbolically calculate the optimal and EER thresholds. Our findings show that in general, the optimal threshold differs from the commonly used EER.

We examine a more complex case, where a mechanism comprises a biometric and an atomic credential (§5). We approximate the user and attacker distribution functions using first and second order approximations, based on the findings of Slobodan et al. [20]. We analyze the mechanism success which reveals that the optimal threshold for a mechanism comprising multiple biometric credentials is not necessarily the same as the optimal threshold if used individually (§5.1). Using numeircal optimization, we observe a reduction of over 66% in the failure rate, when comparing the EER and the optimal threshold. Moreover, we find close form expressions of a mechanism's success when atomic and biometric credentials are integrated and symbolically calculate the optimal and EER thresholds. These expression reveal that the mechanism success can be primarily depandat on an atomic credential or the biometric credential, depands on the atomic probability values. For example, when using *AND mechanism*: The user has full availability to all credentials, while the attacker has limited or no availability; The relation shows that high safe values and low leak values of the atomic credential consider a player as a user even when obtaining a sample with low score value, what makes the mechanism's success mainly dependent on the atomic credential, and vice versa.

Motivated by this observation, we analyze whether the integration of biometric credential with an atomic credential is benificial in terms of mechanism's success. We link between the biometric credential to the improvement it has on the mechanism's success composes an atomic credential and characterize the conditions under which such integration improves success (§6). We prove that when these conditions are not satisfied, it is not worthtile to integrate, as the atomic credential mechanism dominates the integrated mechanism. Furthermore, we show that every biometric credential which is *asymptotic separable*: for every score above or below a specific threshold, the error rates converge to zero; a necessary and sufficient condition, provided that the atomic credential has non-zero probabilities for the states of *safe* and either *loss* or *leak*. We also introduce *asymptotic dominance*: where the genuine distribution dominates the attacker distribution by a factor c from every score above a specific threshold, or the attacker dominates the user by a factor c for every score below a specific threshold. This condition is a sufficient condition and shown as it represent a behavior we expect to observe in biometric credentials.

We quantitatively demonstrate our theoretical analysis with experiments using real public biometric data and a standard algorithm for fingerprint verification [24] (§7). We use a dataset of fingerprints from multiple users and evaluate matching scores by comparing samples of one user against those of other users. When different users attempt to authenticate as one another, the biometric sensors succeed in distinguishing and preventing unintended user access. In this setting, we observe high seperability between the two players, as the algorithm effectively extracts minutiae and distinguishes between different individuals, resulting in low EER. Thus, no significant improvement is observed when comparing the mechanism's success under Equal Error Rate (EER) threshold to the optimal threshold. However, since an attacker does not attempt to authenticate using their own biometric features, the relevant threat model involve attempts to physically impersonate a legitimate user. We conduct a spoofing attack experiment using the LivDet database [7, 14, 17, 25], which includes both genuine fingerprint scans and corresponding spoofing attempts. These spoof samples produce matching scores that closely resemble those of the legitimate user. Integrating the biometric credential into a mechanism comprised of an atomic credential, reduces the mechanism's failure probability by 30-70% compared to the probability with the EER threshold. Furthermore, the results show a 50% reduction in the mechanism's failure probability compared to a mechanism based solely on an atomic credential, demonstrating the benefit of integrating the biometric credential.

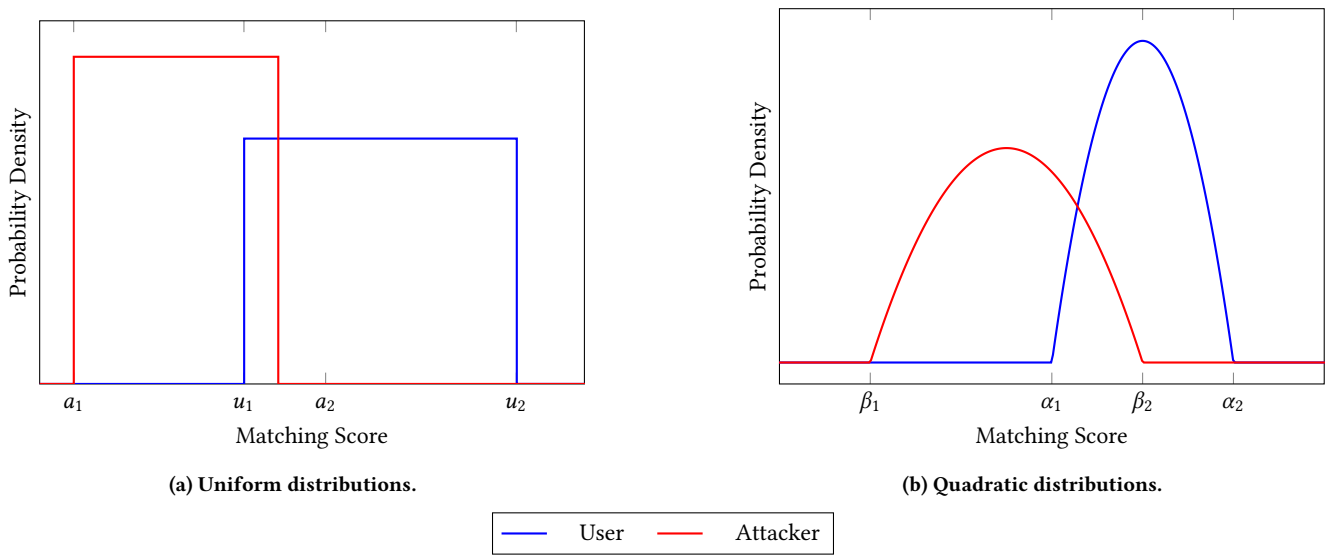
In summary (§8) our contributions are both theoretical and practical: (1) Formalization of the mechanism's success function composed of biometric and atomic credentials; (2) Symbolic analysis of the optimal threshold for biometric and atomic authentication mechanisms. (3) Characterization of the conditions under which biometric credential integration enhances the mechanism's success; (4) Proof that any biometric credential, under a specified condition, always enhances the mechanism's success. (5) Empirical analysis using real biometric data, demonstrating the theoretical analysis. (6) Direct implications to how fingerprint readers should work in practice for immediate security improvements.

2 Related Work

Authentication remains an active and prominent research area, with prior work broadly categorized into two main aspects: credentials and mechanisms. Credentials form the core of the authentication process, commonly manifested as passwords [15], OTPs [19], physical devices [23] or biometric features such as fingerprints, face recognition or iris [10]. An authentication mechanism is a decision making protocol that evaluates a set of credentials provided by a player and output accept or reject [4, 22]. Examples include multi-factor authentication (MFA) [3], an authentication mecahnism composes multiple and different credentials' categories treating each one as an atomic credential.

Extensive work has evaluated and improved the latency and accuracy of biometric systems [5, 7, 9, 12, 14, 17, 25]. Prabhakar et al. [18] examine the tradeoff between two key error metrics of a biometric system, namely False Acceptance Rate (FAR) and False Rejection Rate (FRR). Sarkar et al. [21] introduce the heuristic Equal Error Rate (EER), the point where FAR and FRR are equal, and determine it as an operating point of the biometric system. Hong et al. [8] integrate fingerprint and face recognition to design a fused biometric system that combines the matching scores of both modalities to make a final decision.

Critically, none of these works define an optimizable success criterion for biometric systems. While Hong et al. closely related to our domain, they don't explore the influence of one biometric credential on the other nor do they analyze the integration of biometric credentials along with other credential types in the authenticaion mechanism. Moreover, Sarkar et al. assume a heuristic operating



(a) Uniform distributions.

(b) Quadratic distributions.

Figure 1: Uniform and quadratic user and attacker distributions.

threshold for the biometric system and do not explore how the presence of additional credentials can influence the optimal matching threshold of the biometric system.

Other work has explored authentication mechanism defined as predicate of credential possession. Maram et al. [13] study interactive authentication mechanisms, where the user and attacker interact with the authentication mechanism. Mouallem et al. [16] compare mechanisms' success and aim to identify mechanisms with maximal success when composed of a large number of homogeneous credentials. Eyal [6] analyzes mechanisms modeled as Boolean functions of the credentials. However, these works are partially applicable to our setting. While we adopt their mechanism's success definition and the way they define the credential possession states, they model biometric credentials as atomic credentials. This leaves open the question of whether the integration of biometric credentials along with other credentials, can form the authentication mechanisms to improve its success, which is the focus of our work.

3 Authentication Model

An authentication mechanism M is built on a set of credentials $\{c_1, c_2, \ldots, c_n\}$ and two players: a user U and an attacker A. Both players, try to satisfy the mechanism with their own set of credentials $\{c_1, c_2, \ldots, c_n\}$ in order to reach the asset. Each credential c_i can be either atomic or biometric. In this paper, we will focus on biometric credentials. We will formalize the systems success and identify the optimal operating point for both single and multiple credentials.

3.1 Model Details

We follow the definitions given by Eyal [6]. A mechanism M is defined by a predicate of credential's availability. Each credential can be in one of the following states. *Safe*: Only the user has access to the credential. *Loss*: Neither the user nor the attacker has access to the credential. *Leak*: Both the user and the attacker have access to the credential. *Theft*: Only the attacker has access to the credential.

We define the probability associated with each state as follows: P_{safe}, P_{loss}, P_{leak} and P_{theft}. Furthermore, the matching score produced by the biometric sensor characterizes the genuine distribution, when the sample obtained by the sensor belongs to the real user and the impostor distribution, when the sample obtained by the sensor belongs to an attacker. These distributions are used to establish two error rates: false acceptance rate (FAR), defined as the probability of an impostor being accepted as a genuine individual and false rejection rate (FRR), the probability of a genuine individual being rejected as an impostor.

Denote the credential state by S, i.e., $\{c_1, c_2, \ldots, c_n\} \in S = \{\text{safe}, \text{loss}, \text{leak}, \text{theft}\}$. A scenario σ is defined as a vector of states of each credential in the mechanism. Denote by σ_i the state of credential c_i in scenario σ. An availability vector represents the availablity of all credentials to a user U or an attacker A. Denote by σ^U,σ^A the availability vector of the user and the attacker respectively.

In the following sections we will represent all biometric credentials as the probability function of the matching score s denoted as $P_U[s]$. We follow the definitions given by Jain et al. [10] and generalize it to all biometric credentials. If the stored reference template of the user U is represented by X_U and the acquired input for recognition is represented by X_Q then:

H_0 input X_Q does not come from the same person as the reference template X_U.

H_1 input X_Q comes from the same person as the reference template X_U.

D_O, D_1 are the imposter and geniuine persons trying to authenticate into the account. The matching score s, is usually a single number, that quantifies the similarity between an input X_Q and and the reference store template in the database X_U [18]. If the matching score $s(X_Q, X_U)$ is above a threshold t then decide D_1 else, decide D_0.

4 Single Biometric Credential

For a mechanism M comprised of a single biometric credential c_1, $M^{\text{single}}(c_1) = c_1$ we have built FAR vs FRR curves for Uniform and Quadratic probabilistic distribution functions and defined each one of the states as follows:

$$P_{\text{loss}} = \text{FRR} \cdot (1 - \text{FAR})$$
$$P_{\text{leak}} = \text{FAR} \cdot (1 - \text{FRR})$$
$$P_{\text{theft}} = \text{FRR} \cdot \text{FAR}$$
$$P_{\text{safe}} = 1 - \text{P}_{\text{loss}} - \text{P}_{\text{leak}} - \text{P}_{\text{theft}}$$

The probability for success of a mechanism M, is defined as the probability it accepts legitimate users and rejects attackers. In a single biometric credential mechanism, this is simply the probability that the credential is neither theft, loss or leaked. i.e.,

$$P_{\text{success}}(M^{\text{single}}) = P_{\text{safe}} = 1 - \text{FAR} - \text{FRR} + \text{FAR} \cdot \text{FRR} \tag{1}$$

Until now, it has been common practice to select the operating point at the Equal Error Rate (EER), where FAR = FRR following Sarkar et al. [21]. Our findings indicate that a different operating point can yield a higher P_{safe}, thereby enhancing the mechanism's success. Furthermore, We found a link between the operating point and the variance of the PDFs. We will further explore this in the following sections for different distributions.

4.1 Uniformly Distributed Credential

As a first-order approximation, we model the biometric credential using a uniform distribution. We defined two probability distribution functions for the user and the attacker, P_U and P_A respectively.

$$P_U(s) = \frac{1}{u_2 - u_1} \quad \text{for } u_1 \le s \le u_2$$
$$P_A(s) = \frac{1}{a_2 - a_1} \quad \text{for } a_1 \le s \le a_2$$

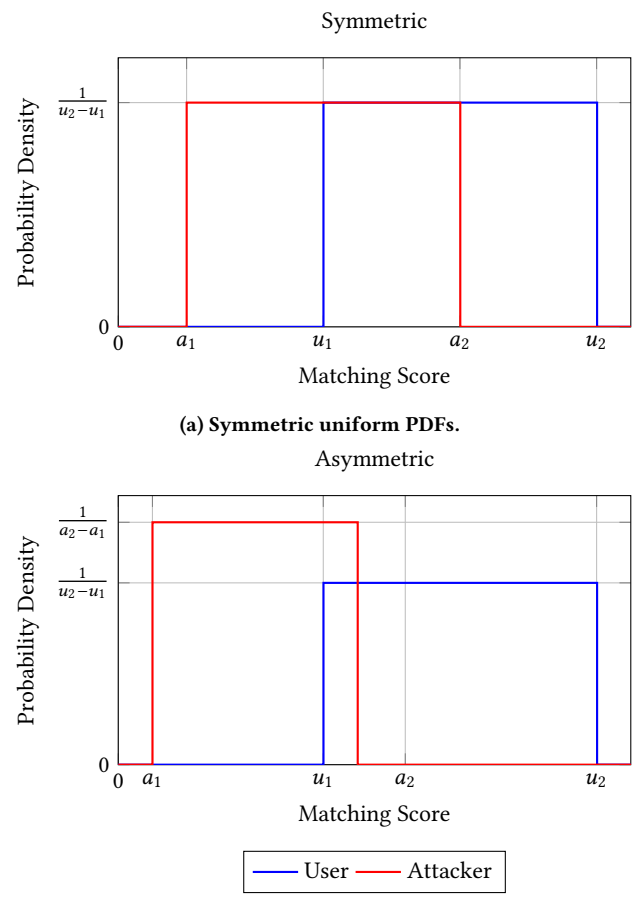
Where u_1, u_2 are the user PDFs' bounds and a_1, a_2 are the attacker PDFs' bounds.

We assume that $a_1 < u_1$ and $a_2 < u_2$ as the x axis represents the matching score s and higher values of s corresponds to the user D_1 as mentioned in the model details. Furthemore, we assume there is an overlaping region between the distributions; otherwise the scenario reduces to a trivial case. We also assume that the user and attacker PDFs are do not fully contain one another, as such cases would render them non-separable. Figure 2 illustrates the case where the user and attacker PDF's are symmetrically and asymmetrically distributed both as function of the matching score s. The probability of success for a single biometric credential mechanism is defined as:

$$P_{\text{success}}(M^{\text{single}}) = P_{\text{safe}}$$
$$P_{\text{success}}(M^{\text{single}}) = 1 - \text{P}_{\text{loss}} - \text{P}_{\text{leak}} - \text{P}_{\text{theft}} = 1 - \text{FRR} - \text{FAR} + \text{FRR} \cdot \text{FAR}$$

Let T denote the threshold. Then the False Acceptance Rate (FAR) and False Rejection Rate (FRR) are defined as:

$$\text{FAR}(T) = \int_T^{a_2} \frac{1}{a_2 - a_1} ds = \begin{cases} 1 & \text{if } T \le a_1 \\ \frac{a_2 - T}{a_2 - a_1} & \text{if } a_1 < T \le a_2 \\ 0 & \text{if } T > a_2 \end{cases}$$



(a) Symmetric uniform PDFs.

(b) Asymmetric uniform PDFs.

Figure 2: Symmetric and asymmetric uniform PDFs.

$$\text{FRR}(T) = \int_{u_1}^{T} \frac{1}{u_2 - u_1} ds = \begin{cases} 0 & \text{if } T \le u_1 \\ \frac{T - u_1}{u_2 - u_1} & \text{if } u_1 < T \le u_2 \\ 1 & \text{if } T > u_2 \end{cases}$$

We define the probability of success as a function of the threshold T by setting the $FAR(T)$ and $FRR(T)$ into the equation and receive:

$$P_{\text{success}}(T) = \begin{cases} 0 & \text{if } T \le a_1 \\ 1 - \frac{a_2 - T}{a_2 - a_1} & \text{if } a_1 < T \le u_1 \\ 1 - \frac{T - u_1}{u_2 - u_1} - \frac{a_2 - T}{a_2 - a_1} + \frac{T - u_1}{u_2 - u_1} \cdot \frac{a_2 - T}{a_2 - a_1} & \text{if } u_1 < T \le a_2 \\ 1 - \frac{T - u_1}{u_2 - u_1} & \text{if } a_2 < T \le u_2 \\ 0 & \text{if } T > u_2 \end{cases}$$

$P_{\text{success}}(T)$ is a continuous function in $[a_1, u_2]$,differentiable in (a_1, u_2) and maintains $P_{\text{success}}(a_1) = P_{\text{success}}(u_2)$ thus by Rolle's Theorem, there exists at least one point in (a_1, u_2) such that:

$$\frac{d}{dT} P_{\text{success}}(T) = 0.$$

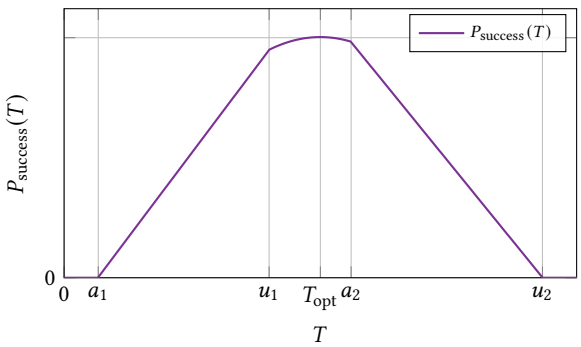


Figure 3: Success probability as a function of the threshold.

The horizontal tangent point is:

$$T = \frac{a_1 + u_2}{2} \quad \text{for } u_1 \leq T \leq a_2$$

$$\frac{d}{dt} P_{\text{success}}(T) > 0, \text{For every } T \in (u_1, \frac{a_1 + u_2}{2})$$

$$\frac{d}{dt} P_{\text{success}}(T) < 0, \text{For every } T \in (\frac{a_1 + u_2}{2}, a_2)$$

Thus, by the First Derivative Test the maximum point is at $T_{\text{Max}} = \frac{a_1+u_2}{2}$.

$$P_{\text{success}}(T_{\text{Max}}) = \frac{(a_1 - u_2)^2}{4(a_1 - a_2)(u_1 - u_2)}$$

We calculate the EER point and show it is not equal to the optimal point:

$$T_{\text{EER}} = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2} \neq \frac{a_1 + u_2}{2} = T_{\text{Max}}$$

The maximum point represnents the optimal threshold of the mechanism. The optimal point indicates that increasing the offset between the two uniform distributions is likely to improve the mechanism's success. Moreover, it suggest an asymmetry between the user's and the attacker's distribution intervals. Figure 3 illustrates the probability of success for a single biometric credential mechanism as function of threshold T.

We can define the FAR and FRR at the optimal point as follows:

$$\text{FAR}(T_{\text{opt}}) = \frac{2a_2 - a_1 - u_2}{2(a_2 - a_1)}$$

$$\text{FRR}(T_{\text{opt}}) = \frac{a_1 + u_2 - 2u_1}{2(u_2 - u_1)}$$

Figure 4 illustrated the FAR vs FRR curve.

4.2 Quadratic Distributed Credential

Our goal is to model credentials whose distributions closely resemble those observed in real measurement data. To this end, we analyze a quadratic probability distribution function, which provides a second order approximation of the biometric credential that can also be seen in real fingerprint measurement data (§7). This choice is also consistent with the findings of Slobodan et al. [20] on Eigenfinger and Eigenpalm feature distributions. Accordingly,

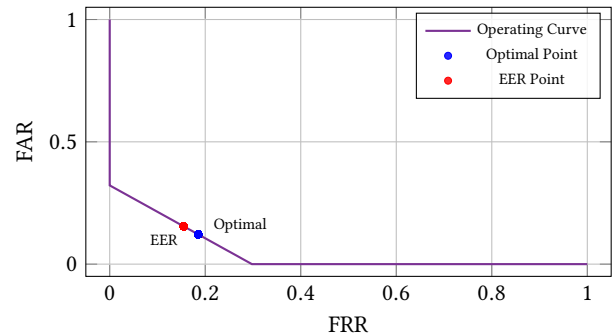


Figure 4: FAR vs. FRR operating curve with optimal and EER points.

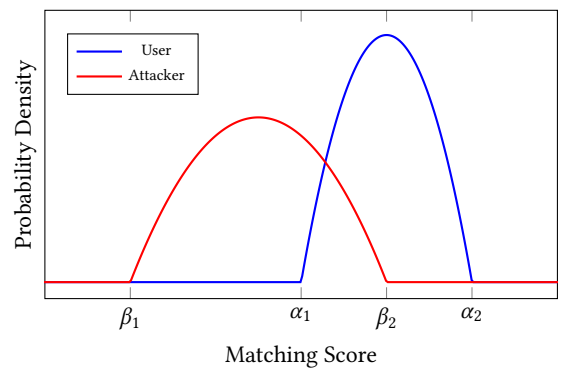


Figure 5: Quadratic user and attacker distributions.

we define two probability density functions (PDFs) for the user and the attacker, denoted by P_U and P_A, respectively.

$$P_U(s) = \begin{cases} a \cdot s^2 + b \cdot s + c & \alpha_1 \leq s \leq \alpha_2 \\ 0 & s > \alpha_2 \vee s < \alpha_1 \end{cases}$$

$$P_A(s) = \begin{cases} d \cdot s^2 + e \cdot s + f & \beta_1 \leq s \leq \beta_2 \\ 0 & s > \beta_2 \vee s < \beta_1 \end{cases}$$

Here, a, b, c and d, e, f denote the quadratic coefficients of the user and attacker distributions, respectively. The parameters α_1, α_2 are the roots of the user's PDF, while β_1, β_2 are the roots of the attacker's PDF. Similarly, to the analysis of the uniform distribution, the x axis represents the matching score, such that higher values of score will be corrspondant to the reference template stored in the database X_U. If $\beta_2 < \alpha_1$, the two distributions do not overlap, and the problem becomes trivial. Conversely, if $\alpha_1 > \beta_1$ and $\beta_2 < \alpha_2$, the attacker's distribution is fully contained within the user's, resulting in a non-separable scenario. We therefore focus on the intermediate and most meaningful configuration, where

$$\beta_1 \leq \alpha_1 \leq \beta_2 \leq \alpha_2,$$

as illustrated in Figure 5.

To reduce the degrees of freedom of the quadratic functions while maintaining a symbolic analysis, we express the coefficients directly in terms of their roots. Under the additional assumption that the integral of each PDF over its roots interval equals 1, the

coefficients become fully determined. Accordingly, we define the coefficients as follows.

$$\alpha_1 \cdot \alpha_2 = \frac{c}{a} \rightarrow a = \frac{c}{\alpha_1 \cdot \alpha_2}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \rightarrow b = -\frac{c \cdot (\alpha_1 + \alpha_2)}{\alpha_1 \cdot \alpha_2}$$

$$\beta_1 \cdot \beta_2 = \frac{f}{d} \rightarrow d = \frac{f}{\beta_1 \cdot \beta_2}$$

$$\beta_1 + \beta_2 = -\frac{e}{d} \rightarrow e = -\frac{f \cdot (\beta_1 + \beta_2)}{\beta_1 \cdot \beta_2}$$

$$\int_{\alpha_1}^{\alpha_2} (a \cdot x^2 + b \cdot x + c) dx = 1 \Rightarrow c = \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3}$$

$$\int_{\beta_1}^{\beta_2} (d \cdot x^2 + e \cdot x + f) dx = 1 \Rightarrow f = \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3}$$

Therefore, we can write the quadratic PDFs as:

$$P_U(s) = \begin{cases} \frac{6}{(\alpha_1 - \alpha_2)^2} \cdot s^2 - \frac{6 \cdot (\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3} \cdot s + \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3} & \alpha_1 \le s \le \alpha_2 \\ 0, & \text{otherwise} \end{cases}$$

$$P_A(s) = \begin{cases} \frac{6}{(\beta_1 - \beta_2)^2} \cdot s^2 - \frac{6 \cdot (\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} \cdot s + \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3} & \beta_1 \le s \le \beta_2 \\ 0, & \text{otherwise} \end{cases}$$

We will find FAR and FRR as a function of the threshold T and define the probability of success based on a single credential mechanism as shown in (§4.1). Define:

$$A(T) = \frac{2(\beta_2^3 - T^3) - 3(\beta_1 + \beta_2)(\beta_2^2 - T^2) + 6\beta_1\beta_2(\beta_2 - T)}{(\beta_1 - \beta_2)^3},$$

$$U(T) = \frac{2(T^3 - \alpha_1^3) - 3(\alpha_1 + \alpha_2)(T^2 - \alpha_1^2) + 6\alpha_1\alpha_2(T - \alpha_1)}{(\alpha_1 - \alpha_2)^3}.$$

Thus,

$$\mathrm{FAR}(T) = \begin{cases} 1, & T \le \beta_1, \\ A(T), & \beta_1 < T \le \beta_2, \\ 0, & T > \beta_2, \end{cases}$$

$$\mathrm{FRR}(T) = \begin{cases} 0, & T \le \alpha_1, \\ U(T), & \alpha_1 < T \le \alpha_2, \\ 1, & T > \alpha_2. \end{cases}$$

$$G(T) = -\frac{(T - \alpha_2)^2 (T - \beta_1)^2 (2T - 3\alpha_1 + \alpha_2)(2T + \beta_1 - 3\beta_2)}{(\alpha_1 - \alpha_2)^3 (\beta_1 - \beta_2)^3}.$$

$$P_{\text{success}}(T) = \begin{cases} 0, & T < \beta_1 \vee T > \alpha_2, \\ \dfrac{(T - \beta_1)^2 (2T + \beta_1 - 3\beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \le T \le \alpha_1, \\ G(T), & \alpha_1 \le T \le \beta_2, \\ -\dfrac{(T - \alpha_2)^2 (2T - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \le T \le \alpha_2. \end{cases}$$

We differentiate $P_{\text{success}}(T)$ and compute its critical points in every interval. By comparing the stationary values obtained in all valid regions, we identify the global maximum, which corresponds to the optimal threshold T_{opt}.

Before writing the piecewise derivative, we define the polynomial coefficients and the following function:

$$q = 4,$$

$$m = -(5\alpha_1 + \alpha_2 + \beta_1 + 5\beta_2),$$

$$n = \alpha_1(3\alpha_2 + \beta_1 + 6\beta_2) - \alpha_2^2 + \alpha_2\beta_2 - \beta_1^2 + 3\beta_1\beta_2,$$

$$\ell = \alpha_1\left(\beta_1^2 - 3\beta_2(\alpha_2 + \beta_1)\right) + \alpha_2^2\beta_2.$$

$$R(T) = -\frac{6(T - \alpha_2)(T - \beta_1)\left(qT^3 + mT^2 + nT + \ell\right)}{(\alpha_1 - \alpha_2)^3 (\beta_1 - \beta_2)^3}.$$

With these coefficients and function $R(T)$, the derivative of $P_{\text{success}}(T)$ is given by the following piecewise expression:

$$\frac{dP_{\text{success}}}{dT} = \begin{cases} 0, & T < \beta_1 \ \vee \ T > \alpha_2, \\ \dfrac{6(T - \beta_1)(T - \beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \le T \le \alpha_1, \\ R(T), & \alpha_1 \le T \le \beta_2, \\ -\dfrac{6(T - \alpha_1)(T - \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \le T \le \alpha_2. \end{cases}$$

In the second interval, the derivative has a single critical point at $T = \beta_1$. Since the derivative is strictly positive for all $T > \beta_1$, this point corresponds to a local minimum rather than a maximum. Similarly, in the fourth interval, the only critical point occurs at $T = \alpha_2$, and the derivative is strictly negative for all $T < \alpha_2$. Therefore, this interval also contains no maximum either.

We therefore turn to the third interval. In this region, the derivative is the product of a cubic polynomial and two linear factors. The linear factors vanish at $T = \beta_1$ and $T = \alpha_2$, both of which lie outside the third interval. Consequently, the existence of critical points within this interval depends solely on the cubic term. Solving this cubic, reveals exactly one real root with the remaining two roots being complex. The closed form of the cubic solution can not be simplified further more than a complex expression. Thus, we denote the solution as T_{cubic}.

By examining the sign of the derivative on both sides of the critical point T_{cubic}, we verify that it constitutes a global maximum of $P_{\text{safe}}(T)$. We denote this maximizing point by T_{opt}.

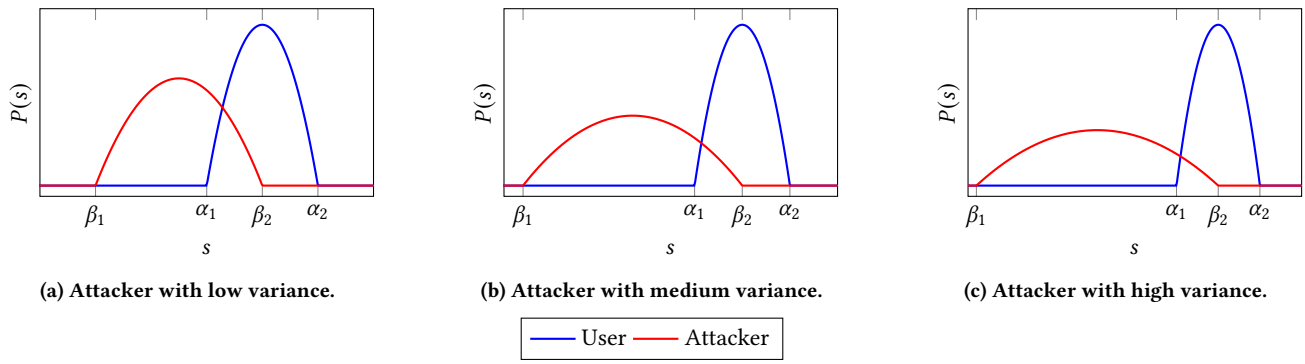
To demonstrate that this optimal threshold differs from the Equal Error Rate (EER) threshold, we computed the EER point separately as follows:

$$T_{\text{EER}} = \frac{\alpha_1\beta_1 - \alpha_2\beta_2}{\alpha_1 - \alpha_2 + \beta_1 - \beta_2} \neq T_{\text{opt}}$$

We illustrate several user and attacker distributions and present their corresponding probability of success curves, expressed as functions of the threshold T in Figure 6 and Figure 7 respectively.

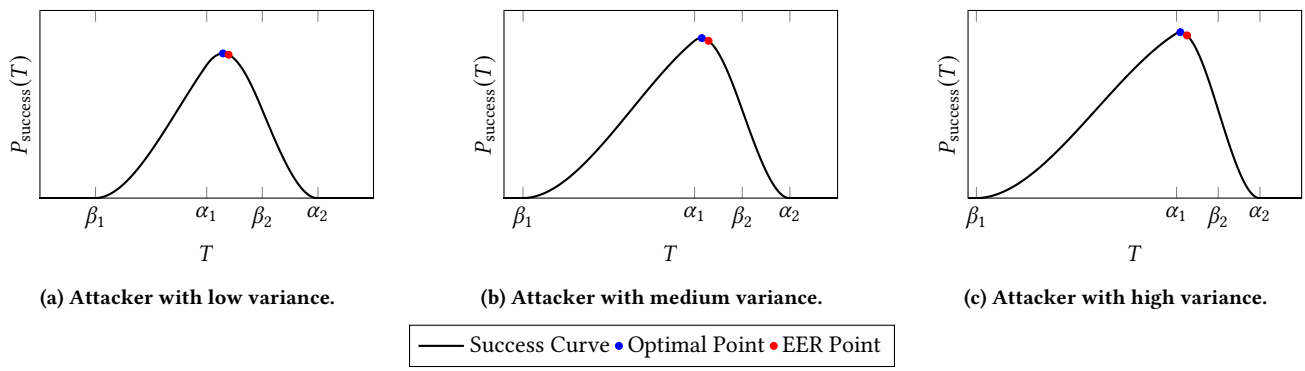
Moreover, we illustrate the FAR–FRR curves for each pair of user and attacker distributions, and mark both the optimal point and the EER point, as shown in Figure 8.

To quantify the effect of the variance on the deviation between the optimal operating point T_{opt} and the equal error rate (EER) point T_{EER}, we numerically computed this gap as a function of the attacker's variance. Decreasing the variance of the attacker achieved by reducing the distance between its roots, corresponds to reduced separability between the two distributions. The simulations



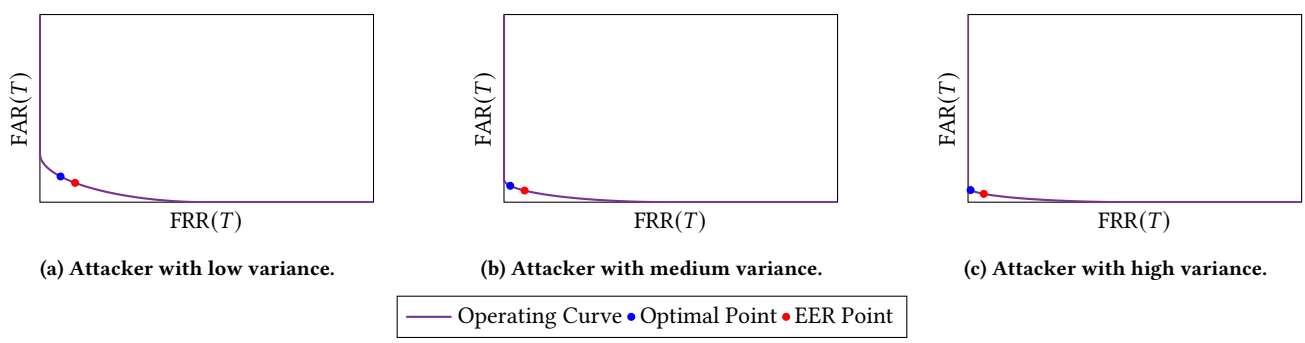
(a) Attacker with low variance. (b) Attacker with medium variance. (c) Attacker with high variance.

Figure 6: Quadratic user and attacker distributions.



(a) Attacker with low variance. (b) Attacker with medium variance. (c) Attacker with high variance.

Figure 7: Success probability for three quadratic attacker distributions.



(a) Attacker with low variance. (b) Attacker with medium variance. (c) Attacker with high variance.

Figure 8: FAR vs. FRR operating curves for different attacker distributions.

reveal that as the variance increases, the gap $|T_{\text{opt}} - T_{\text{EER}}|$ grows accordingly, until the operating threshold moves into a regime where it effectively minimizes only one of the error rates, either FAR or FRR, driving the other to zero. This phenomenon is shown in Figure 9. These findings indicate that higher attacker variance enlarges the potential success advantage gained by operating at T_{opt} rather than at the EER point.

5 Two Credentials

For a mechanism M comprised of two independently and identically distributed credentials, derived from either the same or different PDF's, we defined the integrated system success as the probability of users successfully defend their assets under two scenarios:

a. *AND*: The user has full availability to all credentials, while the attacker has limited or no availability. For successful authentication, we need both credentials to be either in safe state, or one in safe state and the other in leak state. If one of the credentials is either loss ot theft, the owner cannot satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{AND}}) = \mathrm{P}^1_{\text{safe}} \cdot \mathrm{P}^2_{\text{safe}} + \mathrm{P}^1_{\text{safe}} \cdot \mathrm{P}^2_{\text{leak}} + \mathrm{P}^2_{\text{safe}} \cdot \mathrm{P}^1_{\text{leak}} \quad (2)$$

b. *OR*: The user has availability to at least one credential, while the attacker has no availability neither of them. For success, we

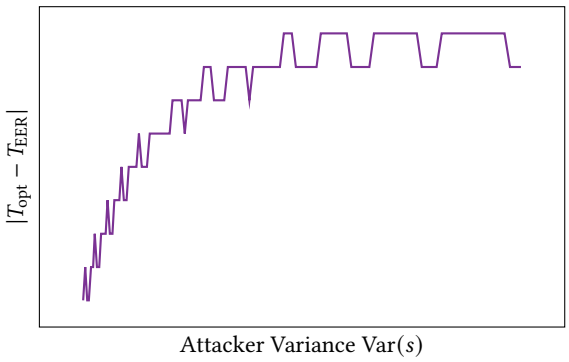


Figure 9: Gap between optimal and EER thresholds as a function of attacker's variance.

need both credential to be either in safe state, or one in safe state and the other in loss state. If one of the credentials is either leak ot theft, the attacker can satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{OR}}) = \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{safe}} + \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{loss}} + \text{P}^2_{\text{safe}} \cdot \text{P}^1_{\text{loss}} \quad (3)$$

For each mechanism type, we analyze the case of two independently and identically distributed credentials in 2 cases:

a. A mechanism comprised of 2 biometric credentials.

b. A mechanism comprised of 1 atomic credential and 1 biometric credential.

We derive the optimal operating point for both *AND* and *OR* mechanisms by maximizing the respective success probabilities as function of the thresholds of each credential.

5.1 2 Biometric Credentials

To analyze the case of a mechanism M comprised of two biometric credentials, we conduct a numerical study over several probability distribution functions. For each distribution pair, we derive the optimal operating point for both *AND* and *OR* mechanisms by maximizing the respective success probabilities with respect to the thresholds of each credential, using a LBFGS optimization algorithm. We additionally compare the optimal operating point to the EER point, thereby characterizing the conditions under which the *AND* or *OR* mechanisms yields superior performance. We started by analyzing the case of Gaussian distributions. We defined the user and attacker distributions as $P_U(s) = \mathcal{N}(\mu_u, \sigma_u^2)$ and $P_A(s) = \mathcal{N}(\mu_a, \sigma_a^2)$ respectively. We set for the first credential $\mu_u = 0.8$, $\sigma_u = 0.12$, $\mu_a = 0.3$ and $\sigma_a = 1$ and for the second credential $\mu_u = 1.0$, $\sigma_u = 0.2$, $\mu_a = 0.4$ and $\sigma_a = 0.18$ as shown in Figure 10.

The optimal operating point for both mechaninsms' types differ from the EER point, yielding higher success probabilities than the EER points. Moreover, we find that the optimal points of both *AND* and *OR* mechanisms are not necessarily the same as the optimal threshold if used individually. We furtherly find that such integration of a biometric credential to a mechanism already comprised of a biometric credential, improves the mechanism success. Hence, integrating two biometric credentials into a mechanism improves its success over using them individually as shown in Figure 11.

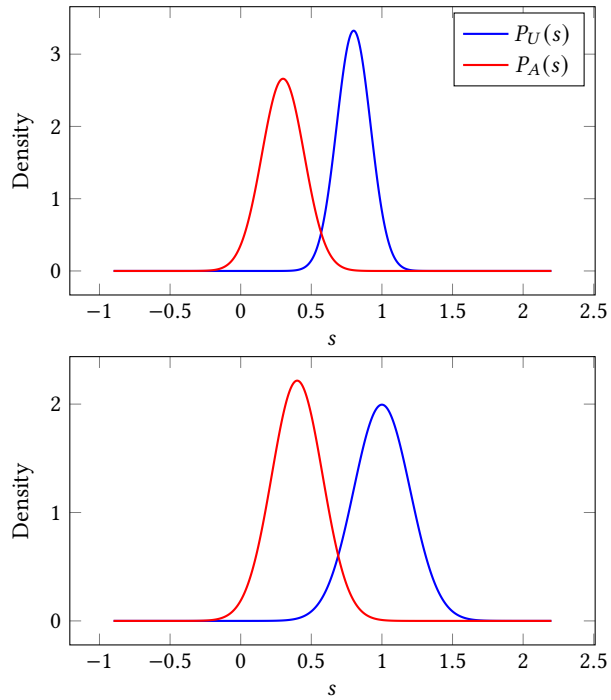


Figure 10: Gaussian user and attacker distributions.

We illustrate the success probability of both *AND* and *OR* mechanisms. We fix a credential to its optimum found out using the LBFGS algorithm and show how the success probability varies as a function of the thresholds of the other credential. The results are shown in Figure 12. In conclusion, all numerical results are summarized in Table 1.

From Table 1, we observe that the optimal operating point yields a higher success probability than the EER point by between 5% to 7%.

5.2 1 Atomic and 1 Biometric Credential

We start by analyzing the case of an *AND* mechanism comprised of 1 atomic credential and 1 biometric credential. Denote the state of credential by s_i, i.e, $i \in (Atomic, Biometric)$, we define the atomic credential with a constant value of each probabilities' states as: $P^1_{safe}, P^1_{loss}, P^1_{leak}, P^1_{theft}$ corrsponding respectively to each one of the states: safe, loss, leak and theft. The biometric credential is defined by its PDFs for the user and the attacker, P_U and P_A respectively. We model the first order approximation of the biometric credential using uniform distribution between the bounds of u_1,u_2 corrsponding to the user distribution and a_1,a_2 corresponds to the attacker, as seen at (§4.1). We derive the probability of the biometric credential as follows:

$$P^2_{safe} = 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T) \cdot \text{FRR}(T),$$
$$P^2_{leak} = \text{FAR}(T)\left(1 - \text{FRR}(T)\right).$$

To analyze the probability of success for the *AND* mechanism, we set the above probabilities into Equation 2. Thus, the probability of success is a function of the threshold T and divided into 5 intervals.

Distribution	$P_{\text{AND}}(T_{\text{EER}})$	$P_{\text{OR}}(T_{\text{EER}})$	$P_{\text{AND}}(T_1^{\text{safe}})$	$P_{\text{OR}}(T_2^{\text{safe}})$	$\boldsymbol{P_{\text{AND}}(T_{\text{opt}})}$	$\boldsymbol{P_{\text{OR}}(T_{\text{opt}})}$
Gaussian	0.910	0.911	0.911	0.911	0.964	0.962
Uniform	0.955	0.955	0.916	0.999	1.000	1.000
Parabolic	0.920	0.921	0.919	0.922	0.987	0.985

Table 1: Mechanism success under Gaussian, Uniform and Parabolic score distributions at three reference operating points: EER-pair $(T_1^{\text{eer}}, T_2^{\text{eer}})$, individual optimum pair $(T_1^{\text{safe}}, T_2^{\text{safe}})$, and joint optima (AND at $(T_1^*, T_2^*)_{\text{AND}}$ and OR at $(T_1^*, T_2^*)_{\text{OR}}$)

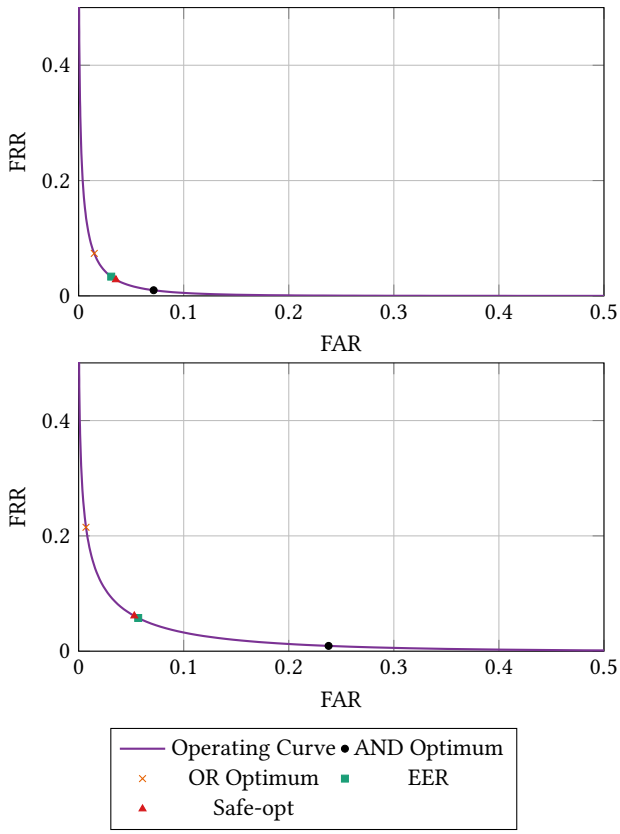
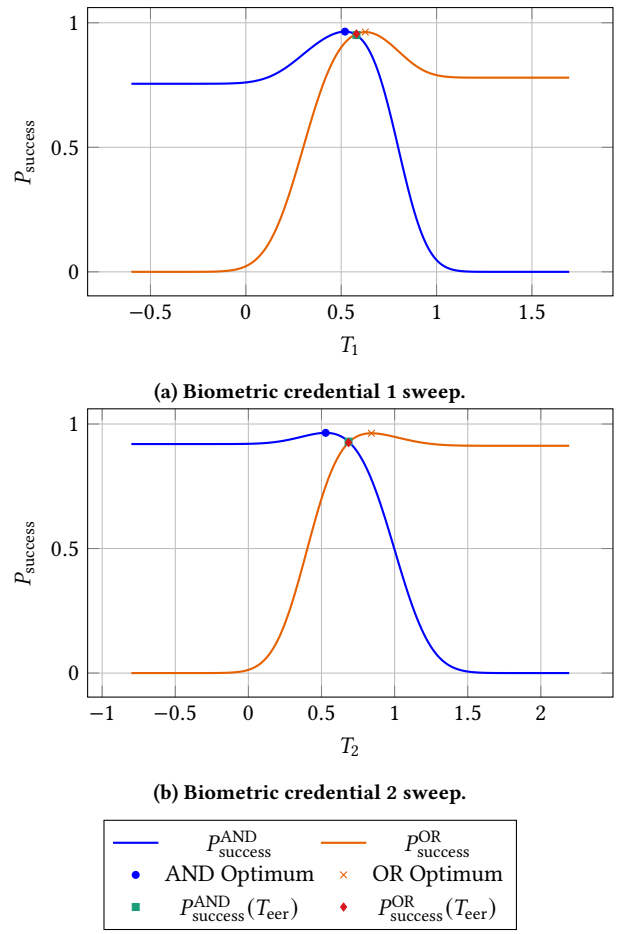


Figure 11: FAR vs. FRR operating curves for two biometric credentials.



(a) Biometric credential 1 sweep.

(b) Biometric credential 2 sweep.

Figure 12: Success probability as a function of the threshold for two biometric credentials.

$$\text{Define: } L(T) = \frac{u_2 - T}{u_2 - u_1}\left[(P^1_{\text{safe}} + P^1_{\text{leak}}) - P^1_{\text{leak}}\frac{a_2 - T}{a_2 - a_1}\right].$$

$$P_{\text{success}}(T) = \begin{cases} P^1_{\text{safe}}, & T \le a_1, \\ (P^1_{\text{safe}} + P^1_{\text{leak}}) - P^1_{\text{leak}}\dfrac{a_2 - T}{a_2 - a_1}, & a_1 < T \le u_1, \\ L(T), & u_1 < T \le a_2, \\ \dfrac{u_2 - T}{u_2 - u_1}(P^1_{\text{safe}} + P^1_{\text{leak}}), & a_2 < T \le u_2, \\ 0, & T > u_2. \end{cases}$$

We analyze the success probability $P_{\text{success}}(T)$ by partitioning the threshold domain into five intervals. Each interval corresponds to a distinct relative positioning of the threshold with respect to the attacker and user distributions.

The first interval, $T \le a_1$, corresponds to the case in which the threshold is lower than the minimum value of the attacker distribution. In this regime, the FAR is equal to 1, while the FRR is equal to 0. Consequently, the safe probability of the biometric

credential is zero, and the success probability converges to P^1_{safe}. Since P^1_{safe} is constant in this interval, no interior maximum exists.

The second interval, $a_1 < T \leq u_1$, corresponds to the case in which the threshold lies between the minimum value of the attacker distribution and the minimum value of the user distribution. In this region, the FAR decreases linearly as a function of T, while the FRR remains equal to 0. As a result, the probabilities of the safe state and the leak state of the biometric credential are $1 - \text{FAR}(T)$ and $\text{FAR}(T)$, respectively. In this interval, $P_{\text{success}}(T)$ is a strictly increasing function, as its derivative is a constant positive value, and therefore the maximum is attained at the boundary point: $T_{\text{opt2}} = u_1$.

The third interval, $u_1 < T \leq a_2$, corresponds to the case in which the threshold lies between the minimum value of the user distribution and the maximum value of the attacker distribution. In this region, both FAR and FRR vary linearly with T, resulting in a success probability that is quadratic as a function of T. Differentiating $P_{\text{success}}(T)$ yields a single critical point inside the interval,

$$T_{\text{opt3}} = \frac{u_2 + a_2 - (a_2 - a_1)\dfrac{P^1_{\text{safe}} + P^1_{\text{leak}}}{P^1_{\text{leak}}}}{2}$$

Since this expression is valid only for the interval $u_1 < T \leq a_2$, the effective maximizer within this interval is obtained by clipping the optimum to the interval boundaries, as follows:

$$T^{(\text{interval})}_{\text{opt},3} = \min\big(a_2,\ \max\big(u_1,\ T_{\text{opt},3}\big)\big)\ .$$

In particular, if $T_{\text{opt},3} < u_1$, the maximum over this interval is attained at $T = u_1$, whereas if $T_{\text{opt},3} > a_2$, the maximum is attained at $T = a_2$. In all configurations consisting of one atomic and such biometric credential, ingertating the mechanism with the biometric credential monotonically enhances its success probability. Evaluating the success probability at the user threshold $T = u_1$ yields

$$P_{\text{success}}(u_1) = P^1_{\text{safe}} + P^1_{\text{leak}}(1 - \frac{a_2 - u_1}{a_2 - a_1}).$$

Since $u_1 > a_1$, it follows that

$$P_{\text{success}}(u_1) > P^1_{\text{safe}},$$

Demonstrating that the integration of the biometric credential strictly improves the success of the mechanism compared to using the atomic credential alone.

By examining the optimal point in this interval under *AND* mechanism, we can deduce that as the variance of the attacker distribution increases, the optimal threshold T_{opt3} shifts toward the lower boundry of the interval, u_1. Moreover, for an atomic credential with high safe values P^1_{safe}, and low leak values P^1_{leak}, the optimal threshold shifts toward the lower boundary as well. Thus, in such case, the success of the mechnaims becomes mainly dependent on the atomic credential. Conversely, an atomic credential with high leak values P^1_{leak} and low safe values P^1_{safe} causes the optimal threshold to shift toward the upper boundary of the interval, a_2. In this scenario, the success of the mechanism relies more heavily on the biometric credential.

The fourth interval, $a_2 < T \leq u_2$, corresponds to the case in which the threshold exceeds the maximum value of the attacker distribution. In this regime, the FAR is equal to 0, while the FRR increases linearly with T. Consequently, $P_{\text{success}}(T)$ is a strictly decreasing function in this interval, and the maximum is attained at the boundary point

$$T_{\text{opt4}} = a_2.$$

Finally, the fifth interval, $T > u_2$, corresponds to the case in which the threshold exceeds the maximum value of the user and the attacker distributions, leaving the success probability equal to 0.

To determine the global optimal operating point, we compare all stationary points and boundary maxima obtained in the valid intervals. The optimal threshold is achieved in the third interval, where the attacker and user distributions overlap and the success function attains an interior maximum. Substituting $T_{\text{opt}} = T_{\text{opt3}}$ into $P_{\text{success}}(T)$ yields

$$P_{\text{success}}(T_{\text{opt}}) = \frac{\left(-a_1\left(P^1_{\text{safe}} + P^1_{\text{leak}}\right) + a_2 P^1_{\text{safe}} + P^1_{\text{leak}}\, u_2\right)^2}{4P^1_{\text{leak}}(a_2 - a_1)(u_2 - u_1)}.$$

We have done similar analysis for the *OR* mechanism. We defined the credentials' probabilities of the mechanism as follows:

$$\begin{aligned}
P_{\text{safe1}} &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\cdot\text{FRR}(T),\\
P_{\text{safe2}} &= k_1,\\
P_{\text{loss1}} &= \text{FRR}(T)\left(1 - \text{FAR}(T)\right),\\
P_{\text{loss2}} &= k_2.
\end{aligned}$$

Therefore, the probability of success for the *OR* mechanism is defined as:

$$P_{\text{success}}(T) = \begin{cases} 0, & T \leq a_1,\\ \dfrac{T - a_1}{a_2 - a_1}\,(k_1 + k_2), & a_1 < T \leq u_1,\\ \dfrac{T - a_1}{a_2 - a_1}\left[k_1 + k_2 - k_2\,\dfrac{T - u_1}{u_2 - u_1}\right], & u_1 < T \leq a_2,\\ k_1 + k_2 - k_2\,\dfrac{T - u_1}{u_2 - u_1}, & a_2 < T \leq u_2,\\ k_1, & T > u_2. \end{cases}$$

By differentiating every interval and identifying the critical points, we found that the optimal threshold that leads to the maximum probability of success lies in the third interval where $u_1 < T \leq a_2$ and is given by:

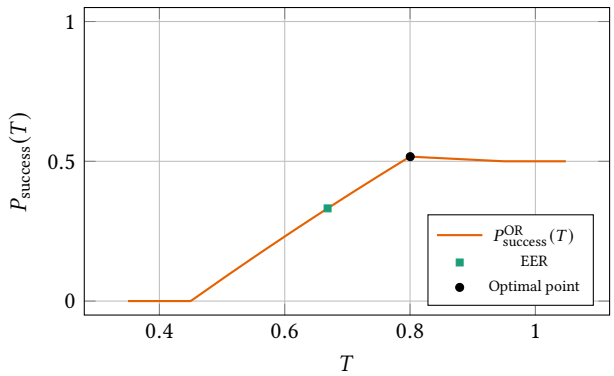
$$T_{\text{opt}} = \frac{a_1 k_2 + u_2(k_1 + k_2) - k_1 u_1}{2k_2}$$

And the maximum probability of success at this point is:

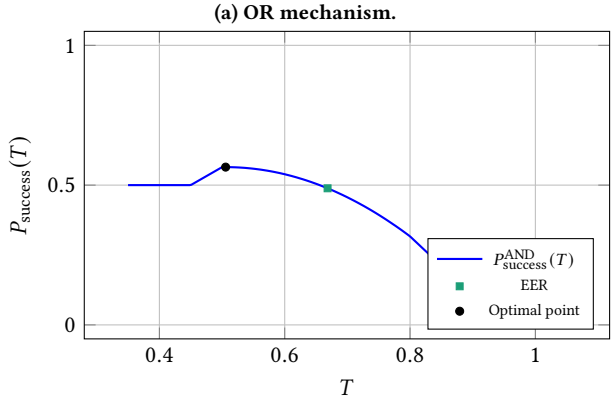
$$P_{\text{success}}(T_{\text{opt}}) = \frac{\left(a_1 k_2 - u_2(k_1 + k_2) + k_1 u_1\right)^2}{4k_2(a_2 - a_1)(u_2 - u_1)}$$

For both mechanisms, we observe that the optimal operating point T_{opt} differs from the equal error rate (EER) point T_{EER}. As the EER depends on the PDF's type solely, we calculate the EER point and show both mechanism's success in Figure 13

$$T_{\text{EER}} = \frac{a_1 u_1 - a_2 u_2}{a_1 - a_2 + u_1 - u_2}$$



(a) OR mechanism.



(b) AND mechanism.

Figure 13: Success probability as a function of the threshold for OR and AND mechanisms composed of an atomic and a biometric credential.

6 Characterizing Success Improvement

We will now characterize the conditions under which integrating a biometric credential into a mechanism composed of an atomic credential can enhance its success. First, we show that for any mechanism whose atomic credential characterized by non-zero probabilities P_{safe} and P_{theft} only, such integration cannot improve mechanism's success (§6.1). Next, we present a necessary and sufficient condition establishing that, for any mechanism composed of an atomic credential with non zero probabilities P_{safe}, P_{leak} or P_{loss}, integrating a biometric credential improves mechanism's success for both finite and infinite distributions (§6.2). Finally, we derive a more commong criterion that provides a sufficient condition under which a biometric credential improves the mechanism's success composed of an atomic credential with non zero probabilities P_{safe}, P_{leak} or P_{loss} (§6.3).

6.1 No Success Gain in the Safe/Theft Case

Proposition 1. *For any mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{theft} only, integrating the mechanism with a biometric credential does not enhance its mechanism's success.*

Proof. Consider a mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{theft} only. In this case, integrating the mechanism with a biometric credential lead to success probability described by Equation 2 and Equation 3. Since $P_{\text{leak}} = P_{\text{loss}} = 0$, the success probability in both mechanism cases reduces to:

$$P_{\text{success}}(T) = P^1_{\text{safe}} \cdot P^2_{\text{safe}}(T)$$

Since $P^2_{\text{safe}}(T) \leq 1$ for every value of T, it follows that $P_{\text{success}}(T) \leq P^1_{\text{safe}}$, and thus integrating a biometric credential does not enhance the success of the mechanism. □

6.2 Sufficient And Necessary Condition for Integrating Biometric Credentials

First, we define the notion of *asymptotic separability* of a biometric credential. Then, we will show that its a necessary and sufficient condition for enhancing the mechanism's success for both finite and infinite distributions.

Definition 1. A biometric credential with user distribution function $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically separable* if

$$\inf_t \min\left\{\frac{\text{FAR}(t)}{1-\text{FRR}(t)}, \frac{\text{FRR}(t)}{1-\text{FAR}(t)}\right\} = 0,$$

where $\text{FAR}(t) = \int_t^{\infty} P_a(s)ds$ and $\text{FRR}(t) = \int_{-\infty}^{t} P_u(s)ds$.

Proposition 2. *For any mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{leak} or P_{loss}, integrating the mechanism with a biometric credential whose user and attacker distributions are asymptotically separable, enhances the success of the mechanism.*

Proof. The success of a mechanism composed of an atomic credential with non-zero probabilities P_{safe} and P_{leak} or P_{loss} is given by P^1_{safe}. A biometric credential is characterized by 4 probability states as function of T: $P^2_{\text{safe}}(T), P^2_{\text{loss}}(T), P^2_{\text{leak}}(T), P^2_{\text{theft}}(T)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned}
P^2_{\text{safe}}(T) &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\text{FRR}(T),\\
P^2_{\text{loss}}(T) &= \text{FRR}(T)\big(1 - \text{FAR}(T)\big),\\
P^2_{\text{leak}}(T) &= \text{FAR}(T)\big(1 - \text{FRR}(T)\big),\\
P^2_{\text{theft}}(T) &= \text{FAR}(T)\text{FRR}(T).
\end{aligned}$$

The success probability of the integrated mechanism comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, by placing the biometric credential in each configuration we can express the success probability of the integrated mechanism as a function of the operating threshold T:

$$P^{\text{AND}}_{\text{success}}(T) = \big(1 - \text{FRR}(T)\big)\Big(P^1_{\text{safe}} + P^1_{\text{leak}}\big(1 - \text{FAR}(T)\big)\Big). \quad (4)$$

$$P^{\text{OR}}_{\text{success}}(T) = \big(1 - \text{FAR}(T)\big)\Big(P^1_{\text{safe}} + P^1_{\text{loss}}\big(1 - \text{FRR}(T)\big)\Big). \quad (5)$$

We need to show that there exists an operating threshold T such that either $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ or $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$. By algebraic

manipulation of Equation 4 we can demand $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ if and only if:

$$\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1-\text{FAR}(T))}$$

Since $P^1_{\text{leak}} \cdot (1-\text{FAR}(T)) \leq P^1_{\text{leak}}$, a sufficient condition for the above inequality is:

$$\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}} \tag{6}$$

Similarly, we can deduce that $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$ if and only if:

$$\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}} \cdot (1-\text{FRR}(T))}$$

Since $P^1_{\text{loss}} \cdot (1-\text{FRR}(T)) \leq P^1_{\text{loss}}$, a sufficient condition for the above inequality is:

$$\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}} \tag{7}$$

By the definition of asymptotic separability, there exists an operating threshold T such that:

$$\inf_t \min\left\{\frac{\text{FAR}(t)}{1-\text{FRR}(t)}, \frac{\text{FRR}(t)}{1-\text{FAR}(t)}\right\} = 0.$$

Equivalently, $\forall \varepsilon > 0, \exists T$ such that either $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \varepsilon$ or $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \varepsilon$. Thus, we can choose ε that equals the right hand side of either Equation 6 or Equation 7 and by definition we know that a T satisfies the inequality exists. More precisely, if

$$\inf_T \frac{\text{FAR}(T)}{1-\text{FRR}(T)} = 0,$$

then the OR mechanism composition is preferable, whereas if

$$\inf_T \frac{\text{FRR}(T)}{1-\text{FAR}(T)} = 0,$$

then the AND mechanism composition is preferable. Therefore, integrating an asymptotically separable biometric credential enhances the success of the mechanism.

□

Proposition 3. *Suppose there $\exists T$ such that the integrated mechanism M composed of biometric credential and atomic credential with non zero probabilities P_{safe} and P_{leak} or P_{loss} achieves a strictly higher success probability than an atomic credential mechanism, i.e.,*

$$P^{integrated}_{success}(T) > P^{atomic}_{success}.$$

Then necessarily, the biometric credential is asymptotically separable.

Proof. We prove by contradiction. Assume for the sake of contradiction, that the biometric credential is not asymptotically separable. Then $\exists \varepsilon > 0$ such that $\forall T$ both $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} \geq \varepsilon$ and $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} \geq \varepsilon$. Since it's given that the integrated mechanism M achieves a strictly higher success probability than an atomic credential mechanism, then there exists a threshold T such that either $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ or $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$. Thus, either Equation 6 or Equation 7 holds for every value of $P^1_{\text{safe}}, P^1_{\text{leak}}, P^1_{\text{loss}}$ in the range of $(0,1)$. In particular, we can choose $P^1_{\text{safe}}, P^1_{\text{leak}}, P^1_{\text{loss}}$ such that either $\frac{P^1_{\text{leak}}}{P^1_{\text{safe}}+P^1_{\text{leak}}} < \varepsilon$ or $\frac{P^1_{\text{loss}}}{P^1_{\text{safe}}+P^1_{\text{loss}}} < \varepsilon$ since the equation holds $\forall \varepsilon > 0$, but both $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} \geq \varepsilon$ and $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} \geq \varepsilon$ derive by the sake of contradiction assumption. Thus, we can deduce that both $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} \geq \frac{P^1_{\text{leak}}}{P^1_{\text{safe}}+P^1_{\text{leak}}}$ and $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} \geq \frac{P^1_{\text{loss}}}{P^1_{\text{safe}}+P^1_{\text{loss}}}$, which contradicts both Equation 6 and Equation 7 under the strictly higher success probability assumption. Therefore, the biometric credential must be asymptotically separable.

□

We show that *asymptotic separability* of the user and attacker distributions is a sufficient and necessary condition for enhancing the success of a mechanism composed of an atomic credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} by integrating it with a biometric credential.

6.3 Common Sufficient Condition

Definition 2. A biometric credential with user distribution fucntion $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically dominated* if:

$$\forall c > 0 \ \exists T \text{ s.t.} : \forall s \geq T : p_a(s) \leq c\, p_u(s)$$

or

$$\forall c > 0 \ \exists T \text{ s.t.} : \forall s \leq T : p_u(s) \leq c\, p_a(s)$$

We will show that *asymptotically dominancy* of user and attacker distribution functions is a sufficient condition for enhancing an atomic credential mechanism's success.

Proposition 4. *For any mechanism M composed of an atomic credential and success function $P^{atomic}_{success}$, integrating the mechanism with an asymptotically dominated biometric credential improves its achievable success. That is,*

$$P^{integrated}_{success}(T) > P^{atomic}_{success}.$$

Proof. An atomic credential is characterized by fixed probability states $P^1_{\text{safe}}, P^1_{\text{loss}}, P^1_{\text{leak}}, P^1_{\text{theft}}$. Thus the success probability of a mechanism composed solely of this atomic credential is constant and given by:

$$P^{\text{atomic}}_{\text{success}} = P^1_{\text{safe}}$$

A biometric credential is characterized by 4 probability states that depend on the operating threshold T: $P^2_{\text{safe}}(T)$, $P^2_{\text{loss}}(T)$, $P^2_{\text{leak}}(T)$, $P^2_{\text{theft}}(T)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned}
P^2_{\text{safe}}(T) &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\text{FRR}(T),\\
P^2_{\text{loss}}(T) &= \text{FRR}(T)\big(1-\text{FAR}(T)\big),\\
P^2_{\text{leak}}(T) &= \text{FAR}(T)\big(1-\text{FRR}(T)\big),\\
P^2_{\text{theft}}(T) &= \text{FAR}(T)\text{FRR}(T).
\end{aligned}$$

The success probability of the integrated mechanism comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, we can express the success probability of the integrated mechanism as a function of the operating threshold T as shown in Equation 4 and Equation 5. We previously observed that a sufficient condition for improvement of mechanism's success is described by Equation 6

and Equation 7. Thus, we need to show that there exists an operating threshold T such that either:

$$\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}}$$

or

$$\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}}$$

We expand left hand side of Equation 6.

$$\frac{\text{FRR}(T)}{1-\text{FAR}(T)} = \frac{\int_{-\infty}^{T} P_u(s)ds}{\int_{-\infty}^{T} P_a(s)ds}$$

By Definition 2, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold T such that for every $s \leq T$, $P_u(s) \leq c \cdot P_a(s)$. Therefore:

$$\frac{\int_{-\infty}^{T} P_u(s)ds}{\int_{-\infty}^{T} P_a(s)ds} \leq \frac{\int_{-\infty}^{T} c \cdot P_a(s)ds}{\int_{-\infty}^{T} P_a(s)ds} = c$$

This is true $\forall c > 0$, thus we can choose c such that satisfies Equation 6.

We expand left hand side of Equation 7.

$$\frac{\text{FAR}(T)}{1-\text{FRR}(T)} = \frac{\int_{T}^{\infty} P_a(s)ds}{\int_{T}^{\infty} P_u(s)ds}$$

By Definition 2, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold T such that for every $s \geq T$, $P_a(s) \leq c \cdot P_u(s)$. Therefore:

$$\frac{\int_{T}^{\infty} P_a(s)ds}{\int_{T}^{\infty} P_u(s)ds} \leq \frac{\int_{T}^{\infty} c \cdot P_u(s)ds}{\int_{T}^{\infty} P_u(s)ds} = c, \forall c > 0$$

We showed that $\forall c > 0$ either $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < c$ or $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < c$. Therefore, integrating an asymptotically dominated biometric credential with an atomic credential improves mechanism success. □

This proof is not trivial, as the integration of another atomic credential to an atomic credential mechanism, does not always enhance its success. For example, if the mechanism is comprised of 1 atomic credential with high safe values and low leak values, adding an atomic credential with low safe values and high leak values would decrease the overall mechanism's success. Thus, Proposition 4 shows that by using an *asymptotically dominated* biometric credential, we can always improve the success of all mechanism's composed of an atomic credential . Moreover, *asymptotically dominancy* condition supports practical scenarios, as we expect that for very low values of score s the attacker distribution dominates the user, vice versa for very high values of s. We supports this claim in with real public biometric data in Section 7.

7 Measuremens

We now present measurements based on fingerprints taken from FVC2002 database [1] and FVC2004 database [2]. We used a standard fingerprint verification algorithm SourceAFIS created by Vazan [24] for scoring fingerprints out of their images based on minutiae triplets. We chose to use fingerprints, as it is a widely common and used example of a biometric credential in authentication mechanisms. For a test set, we took a template out of FVC2002 database

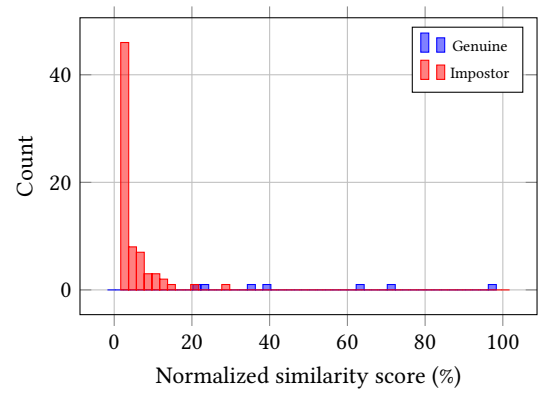


Figure 14: Score histogram for genuine and impostor comparisons.

1 of user 103 and compare them against the whole database of both FVC2002 and FVC2004. A geniune will be considered the template of user 103 against its other templates while imposter considered as all the other combinations. For scoring methodology, we define the final score as the maximum similarity across all 5 user templates. For example, if the matching scores between a sample and the 5 probes are [0.1,0.3,0.6,0.9,0.4], then the resulting score is 0.9. We maintain same approach as Jain et al [9] and Cappelli et al [5] and first built histogram of the geniune and imposter combinations as shown in Figure 14. Then, we turn the histogram into a smoothed probability distribution function. We find the False rejection rate and False acceptance rate and calculate the Equal Error Rate (EER) as shown in Figure 15. The most important outcome from this approach was to achieve similar EER rate of around 2% as achieved from Maio et al. [12], to objectively make a comparison with state-of-the-art algorithms. We calculate the probability success function as given in Equation 1 of a mechanism composed of a biometric credential and show that the optimal operating point is different from the EER point and achieves higher success probability as shown in Figure 16. Finally, we illustrate the probability success function of a mechanism composed of an atomic credential with $P_{\text{safe}} = 0.9$ and $P_{\text{leak}} = 0.04$ $P_{\text{loss}} = 0.04$ and integrate it with the biometric credential.

We show that the optimal operating point of the integrated mechanism is different from the EER point and achieves higher success probability by around 3% compared to the EER point, as the EER itself is only 1.8%. In addition, we can observe that the optimal point in both mechanisms achieves higher success probability than both the atomic credetial mechanisms and biometric credential mechanism when tested individually as shown in Figure 16. Although there is an improvement in the success probability of the integrated mechanism, we can observe the improvement is not as significant as the one observed in the theoretical analysis. This can be explained by the fact that the EER point of the biometric credential is already very low, indicating high seperability between a user and an attacker and thus, there is no much room for improvement. In that scenario, we compared several templates of user 103 against templates of different users. However, our main approach in this paper was to increase the mechanism's success in scenarios where an

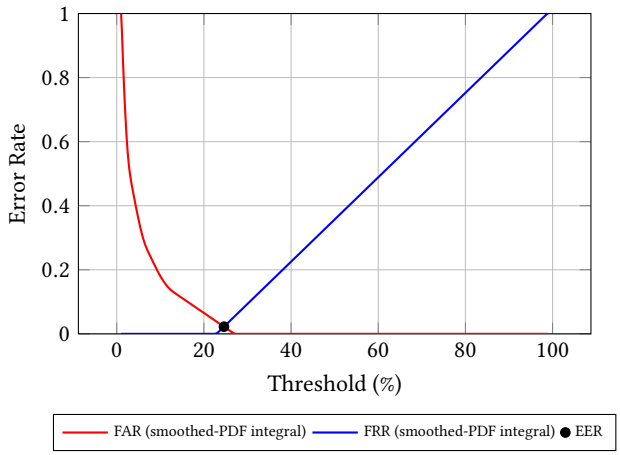


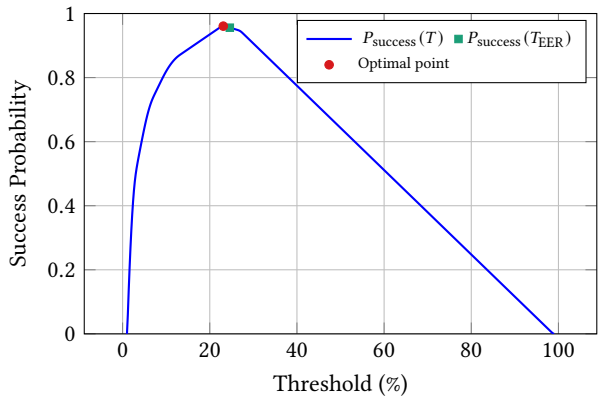
Figure 15: FAR and FRR as a function of the decision threshold, with the Equal Error Rate (EER) point.

imposter will try to spoof the user templates and authenticate into the user account. Hence, we expect the imposter to use spoofing tools, producing samples resamble with score distribution function that much more similar to the genuine distribution function, which might lead to increased EER point.

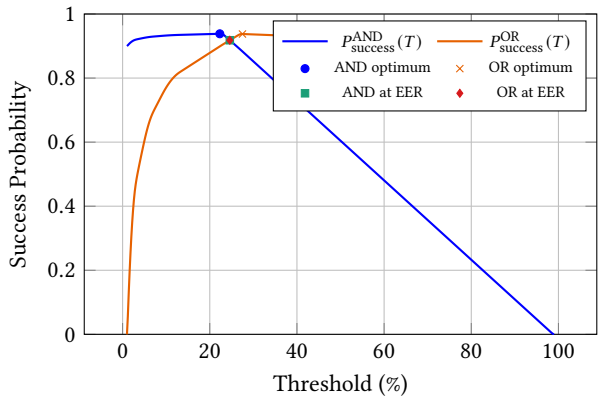
To support this claim, we used the LivDet2015 database [7, 14, 17, 25], which contains a large-scale database of live and spoof fingerprint samples. We follow the same methodology as before and used the SourceAFIS algorithm [24]. As probe templates, we selected six Digital Persona scanner images of user 002_0. Each probe was matched against 14 additional images of the same user to generate genuine scores. For each target image, we define the final score methodolodgy as the maximum similarity across 6 probe templates. For example, if the matching scores between a target image and the six probes are [0.3, 0.5, 0.4, 0.8, 0.9, 0.7], then the resulting score is 0.9. To simulate a spoofing attack, we used fake fingerprint images of the same user (002_0) produced by Gelatine, Woodglue, Latex and Ecoflex and match them against the probe templates to obtain the imposter score distribution. The same maximum-score aggregation was applied in this case as well. We present the resulting histogram in Figure 17. We then compute the EER point and plot the FAR-FRR curve, as shown in Figure 18. As expected, the EER increased significantly to 17.8%. Finally, we evaluated the success probability for both a single biometric credential and for a mechanism composed of an atomic credential and a biometric credential, as shown in Figure 19. We can observe that the optimal operating point reducing the failure probability of an AND mechanism by 60% and Or mechanism by 44% compared to the EER point. Moreover, we reduced the failure probability by 50% when integrating into AND mechanism compared to an atomic credential mechanism with $P_{\text{safe}} = 0.8, P_{\text{loss}} = 0.04, P_{\text{leak}} = 0.15, P_{\text{theft}} = 0.01$ probabilities.

8 Conclusion

We formalize the authentication mechanism model consisting, for the first time, of both biometric and atomic credential mechanisms. We analyze the success of each mechanism type and present a new



(a) Single biometric credential mechanism's success.



(b) biometric credential integrated with an atomic credential into AND and OR mechanisms.

Figure 16: Success probability as a function of the threshold. The AND and OR mechanisms achieve success probabilities of 0.93 **and** 0.94 **at the EER point, and** 0.96 **and** 0.95 **at their respective optimal operating points.**

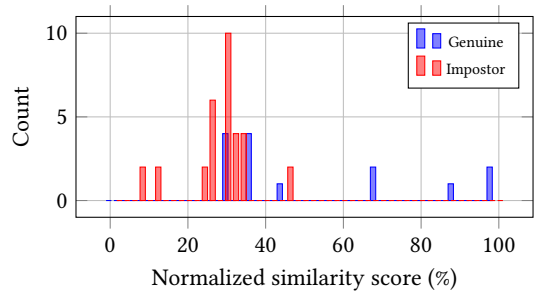


Figure 17: LivDet score histogram using the best score over probe templates.

operating point that achieves higher success probability by modelling each credential with four probability states. We show every mechanism comprised of atomic credentials can be enhanced by integrating it with asymptotically separable biometric credential.

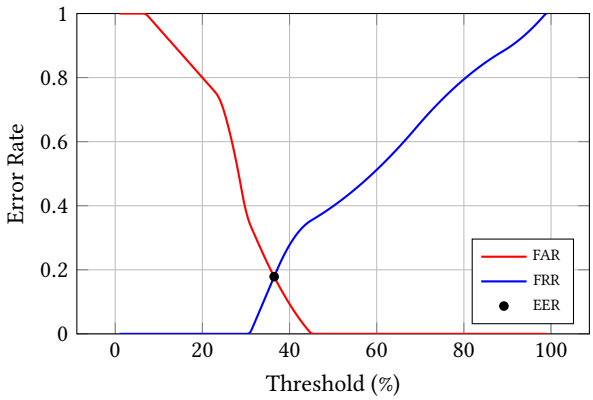
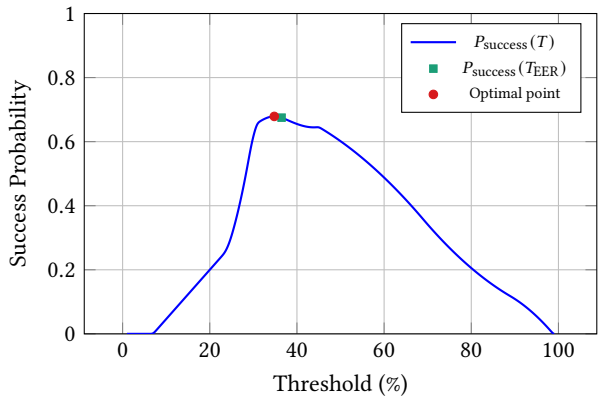
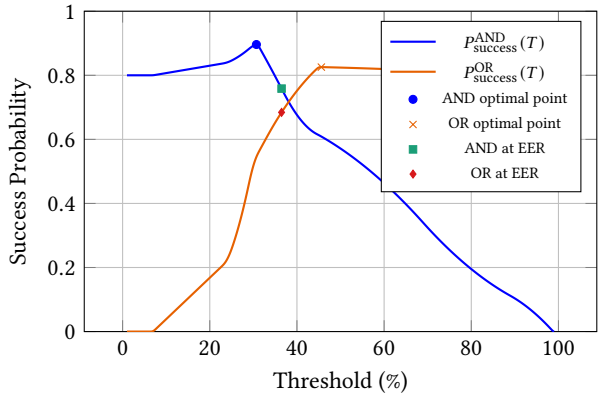


Figure 18: FAR and FRR as a function of the threshold using the best score over probe templates. The Equal Error Rate (EER) occurs at ≈ 17.8%.

Our analysis shows significant reduction of the failure probability under spoofing attempts, which became prevalant and easier on biometric systems in today's artificial intelligence era. Building new mechanisms, based on the suggested dynamic operating point done (§4) is a promising direction for future work. Our findings have important and impactful implications for the design of secure authentication biometric systems, as we suggest letting the integrators to set the thresholds of the biometric sensors. Harnessing our approach, amy affect millions of users around the world to enhace their security of digital and physical assets.



(a) Single biometric credential.



(b) AND and OR mechanisms.

Figure 19: Success probability as a function of the decision threshold using the best score over probe templates. The AND mechanism achieves a maximal success probability of 0.896**, while the OR mechanism achieves** 0.826**. At the EER operating point, the success probabilities are** 0.759 **and** 0.684 **for the AND and OR mechanisms, respectively.**

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- Enumerates all artifacts needed to evaluate the paper's core contributions (e.g., code, datasets, models, configuration files, scripts, documentation, benchmarks).
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- Explicitly justifies any artifact that cannot be shared (e.g., due to licensing restrictions, responsible disclosure concerns, safety or privacy of study subjects, or deployment risks if adversarial methods are released prematurely). When full sharing is not possible, authors are encouraged to provide partial, synthetic, or redacted artifacts that still allow reviewers to assess the methodology.
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